

Exercise 4A**Solution 1:**

(i) $3x + 5y = 7.5$

$$\Rightarrow 3x + 5y - 7.5 = 0$$

$$\Rightarrow 3x + 5y - \frac{15}{2} = 0$$

$$\Rightarrow 6x + 10y - 15 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 6, b = 10 \text{ and } c = -15$$

(ii) $2x - \frac{y}{5} + 6 = 0$

$$\Rightarrow 10x - y + 30 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 10, b = -1 \text{ and } c = 30$$

(iii) $3y - 2x = 6$

$$\Rightarrow -2x + 3y - 6 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = -2, \quad b = 3 \text{ and } c = -6$$

(iv) $4x = 5y$

$$\Rightarrow 4x - 5y = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 4, \quad b = -5 \text{ and } c = 0$$

(v) $\frac{x}{5} - \frac{y}{6} = 1$

$$\Rightarrow \frac{6x - 5y}{30} = 1$$

$$\Rightarrow 6x - 5y = 30$$

$$\Rightarrow 6x - 5y - 30 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 6, b = -5 \text{ and } c = -30$$

$$\text{(vi)} \quad \sqrt{2}x + \sqrt{3}y = 5$$

$$\Rightarrow \sqrt{2}x + \sqrt{3}y - 5 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = \sqrt{2}, b = \sqrt{3} \text{ and } c = -5$$

Solution 2:

$$\text{(i)} \quad x = 6$$

$$\Rightarrow x - 6 = 0$$

$$\Rightarrow 1x + 0y - 6 = 0$$

$$\Rightarrow x + 0y - 6 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 1, b = 0 \text{ and } c = -6$$

$$\text{(ii)} \quad 3x - y = x - 1$$

$$\Rightarrow 3x - x - y + 1 = 0$$

$$\Rightarrow 2x - y + 1 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 2, \quad b = -1 \text{ and } c = 1$$

$$\text{(iii)} \quad 2x + 9 = 0$$

$$\Rightarrow 2x + 0y + 9 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 2, b = 0 \text{ and } c = 9$$

$$\text{(iv)} \quad 4y = 7$$

$$\Rightarrow 0x + 4y - 7 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 0, b = 4 \text{ and } c = -7$$

$$\text{(v)} \quad x + y = 4$$

$$\Rightarrow x + y - 4 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 1, b = 1 \text{ and } c = -4$$

$$\text{(vi)} \quad \frac{x}{2} - \frac{y}{3} = \frac{1}{6} + y$$

$$\Rightarrow \frac{x}{2} - \frac{y}{3} - y = \frac{1}{6}$$

$$\Rightarrow \frac{3x - 2y - 6y}{6} = \frac{1}{6}$$

$$\Rightarrow 3x - 8y = 1$$

$$\Rightarrow 3x - 8y - 1 = 0$$

On comparing this equation with $ax + by + c = 0$, we obtain

$$a = 3, b = -8 \text{ and } c = -1$$

Solution 3:

Given equation is $5x - 4y = 20$ (1)

$$\text{(i)} \quad (4, 0)$$

Putting $x = 4$ and $y = 0$ in L.H.S. of given equation (1), we get

$$\text{L. H. S.} = 5x - 4y$$

$$= 5 \times 4 - 4 \times 0$$

$$= 20 - 0$$

$$= 20$$

$$= \text{R. H. S.}$$

Hence, $(4, 0)$ is the solution of the given equation.

(ii) (0, 5)

Putting $x = 0$ and $y = 5$ in L.H.S. of given equation (1), we get

$$\text{L.H.S.} = 5x - 4y$$

$$= 5 \times 0 - 4 \times 5$$

$$= 0 - 20$$

$$= -20$$

$$\neq \text{R.H.S.}$$

Hence, (0, 5) is not the solution of the given equation.

(iii) $\left(-2, \frac{5}{2}\right)$

Putting $x = -2$ and $y = \frac{5}{2}$ in L. H. S. of given equation (1), we get

$$\text{L.H.S.} = 5x - 4y$$

$$= 5 \times (-2) - 4 \times \frac{5}{2}$$

$$= -10 - 10$$

$$= -20$$

$$\neq \text{R.H.S.}$$

Hence, $\left(-2, \frac{5}{2}\right)$ is not the solution of the given equation.

(iv) (0, -5)

Putting $x = 0$ and $y = -5$ in L.H.S. of given equation (1), we get

$$\text{L.H.S.} = 5x - 4y$$

$$= 5 \times 0 - 4 \times (-5)$$

$$= 0 + 20$$

$$= 20$$

$$= \text{R.H.S.}$$

Hence, (0, -5) is the solution of the given equation.

$$(v) \left(2, -\frac{5}{2}\right)$$

Putting $x = 2$ and $y = -\frac{5}{2}$ in L. H. S. of given equation (1), we get

$$\text{L.H.S.} = 5x - 4y$$

$$= 5 \times 2 - 4 \times \left(-\frac{5}{2}\right)$$

$$= 10 + 10$$

$$= 20$$

$$= \text{R.H.S.}$$

Hence, $\left(2, -\frac{5}{2}\right)$ is the solution of the given equation.

Solution 4:

$$(a) 2x - 3y = 6 \quad \dots (1)$$

Substituting $x = 0$ in equation (1), we get

$$2(0) - 3y = 6$$

$$\Rightarrow 0 - 3y = 6$$

$$\Rightarrow 3y = -6$$

$$\Rightarrow y = -2$$

So, $(0, -2)$ is the solution of the equation.

Substituting $y = 0$ in the equation (1), we get

$$2x - 3 \times 0 = 6$$

$$\Rightarrow 2x - 0 = 6$$

$$\Rightarrow 2x = 6$$

$$\Rightarrow x = 3$$

So, $(3, 0)$ is the solution of the equation.

Substituting $x = 1$ in the equation (1), we get

$$2 \times 1 - 3y = 6$$

$$\Rightarrow 2 - 3y = 6$$

$$\Rightarrow 3y = 2 - 6$$

$$\Rightarrow y = -\frac{4}{3}$$

So, $\left(1, -\frac{4}{3}\right)$ is the solution of the equation.

Substituting $y = 2$ in the equation (1), we get

$$2x - 3 \times 2 = 6$$

$$\Rightarrow 2x - 6 = 6$$

$$\Rightarrow 2x = 12$$

$$\Rightarrow x = 6$$

So, $(6, 2)$ is the solution of the equation.

Substituting $y = 4$ in the equation (1), we get

$$2x - 3 \times 4 = 6$$

$$\Rightarrow 2x - 12 = 6$$

$$\Rightarrow 2x = 18$$

$$\Rightarrow x = 9$$

So, $(9, 4)$ is the solution of the given equation.

Substituting $x = -3$ in the given equation, we get

$$2 \times (-3) - 3y = 6$$

$$\Rightarrow -6 - 3y = 6$$

$$\Rightarrow 3y = -12$$

$$\Rightarrow y = -4$$

So, $(-3, -4)$ is the solution of the given equation.

$\therefore$ Five solutions are

x	0	3	6	9	-3
y	-2	0	2	4	-4

$$(b) \frac{2x}{5} + \frac{3y}{10} = 3 \quad \dots (1)$$

Putting $x = 0$ in equation (1)

$$2 \times \frac{0}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow 3y = 30$$

$$\Rightarrow y = 10$$

So, $(0, 10)$ is the solution of the given equation.

Putting $x = 3$ in equation (1), we get

$$\frac{2 \times 3}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow \frac{6}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow \frac{6}{5} \times 10 + \frac{3y}{10} \times 10 = 3 \times 10$$

$$\Rightarrow 12 + 3y = 30$$

$$\Rightarrow 3y = 18$$

$$\Rightarrow y = 6$$

So, $(3, 6)$ is the solution of the given equation.

Putting $x = -3$ in equation (1), we get

$$\frac{2 \times (-3)}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow 4(-3) + 3y = 30 \quad [\text{Multiplying both sides by 10}]$$

$$\Rightarrow -12 + 3y = 30$$

$$\Rightarrow 3y = 42$$

$$\Rightarrow y = 14$$

So, $(-3, 14)$ is the solution of the given equation.

Putting $x = 6$ in equation (1), we get

$$\frac{2 \times 6}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow \frac{12}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow \frac{24 + 3y}{10} = 3$$

$$\Rightarrow 24 + 3y = 30$$

$$\Rightarrow 3y = 30 - 24$$

$$\Rightarrow 3y = 6$$

$$\Rightarrow y = 2$$

So, (6, 2) is the solution of the given equation.

Putting $x = 9$ in equation (1), we get

$$\frac{2 \times 9}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow \frac{18}{5} + \frac{3y}{10} = 3$$

$$\Rightarrow \frac{36 + 3y}{10} = 3$$

$$\Rightarrow 36 + 3y = 30$$

$$\Rightarrow 3y = 30 - 36$$

$$\Rightarrow 3y = -6$$

$$\Rightarrow y = -2$$

So, (9, -2) is the solution of the given equation.

$\therefore$ Five solutions are

x	0	3	-3	6	9
y	10	6	14	2	-2

(c) $3y = 4x$... (1)

Putting $x = 3$ in equation (1), we get

$$3y = 4 \times 3$$

$$\Rightarrow 3y = 12$$

$$\Rightarrow y = 4$$

So, $(3, 4)$ is the solution of the given equation.

Putting $x = -3$ in equation (1), we get

$$3y = 4 \times (-3)$$

$$\Rightarrow 3y = -12$$

$$\Rightarrow y = -4$$

So, $(-3, -4)$ is the solution of the given equation.

Putting $x = -6$ in equation (1), we get

$$3y = 4 \times (-6)$$

$$\Rightarrow 3y = -24$$

$$\Rightarrow y = -8$$

So, $(-6, -8)$ is the solution of the given equation.

Putting $y = 8$ in equation (1), we get

$$3 \times 8 = 4x$$

$$\Rightarrow 4x = 24$$

$$\Rightarrow x = 6$$

So, $(6, 8)$ is the solution of the given equation.

Putting $x = 9$ in equation (1), we get

$$3y = 4 \times 9$$

$$\Rightarrow 3y = 36$$

$$\Rightarrow y = 12$$

So, $(9, 12)$ is the solution of the given equation.

$\therefore$ Five solutions are

x	3	-3	-6	6	9
y	4	-4	-8	8	12

Solution 5:

$$5x - 3y = k, \dots (i)$$

Since $x = 3$ and $y = 4$ is a solution of the equation (1) so, it will satisfy the equation.

Now, Putting $x = 3$ and $y = 4$ in equation (i)

$$5 \times 3 - 3 \times 4 = k$$

$$\Rightarrow 15 - 12 = k$$

$$\Rightarrow k = 3$$

Solution 6:

$$4x - 3y + 1 = 0 \dots (1)$$

Since $x = 3k + 2$ and $y = 2k - 1$ is a solution of the equation (1) so, it will satisfy the equation.

Now, Putting $x = 3k + 2$ and $y = 2k - 1$ in equation (1)

$$4 \times (3k + 2) - 3 \times (2k - 1) + 1 = 0$$

$$\Rightarrow 12k + 8 - 6k + 3 + 1 = 0$$

$$\Rightarrow 6k + 12 = 0$$

$$\Rightarrow 6k = -12$$

$$\Rightarrow k = -2$$

Solution 7:

Let the cost of one pencil be Rs. x and that of one ballpoint be Rs. y .

$$\text{Cost of 5 pencils} = 5x$$

$$\text{Cost of 2 ballpoints} = 2y$$

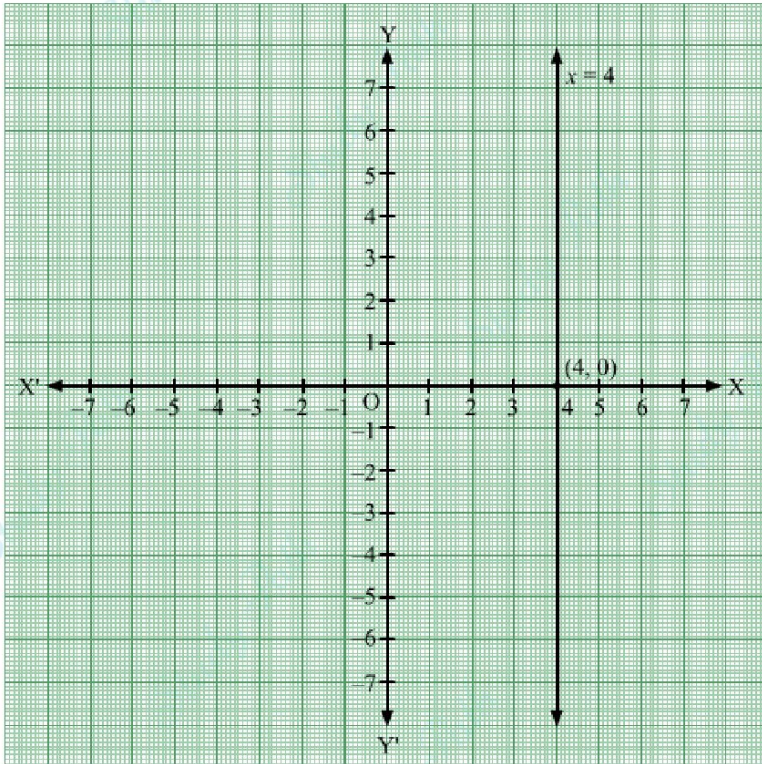
$$\text{Cost of 5 pencils} = \text{cost of 2 ballpoints}$$

$$\Rightarrow 5x = 2y$$

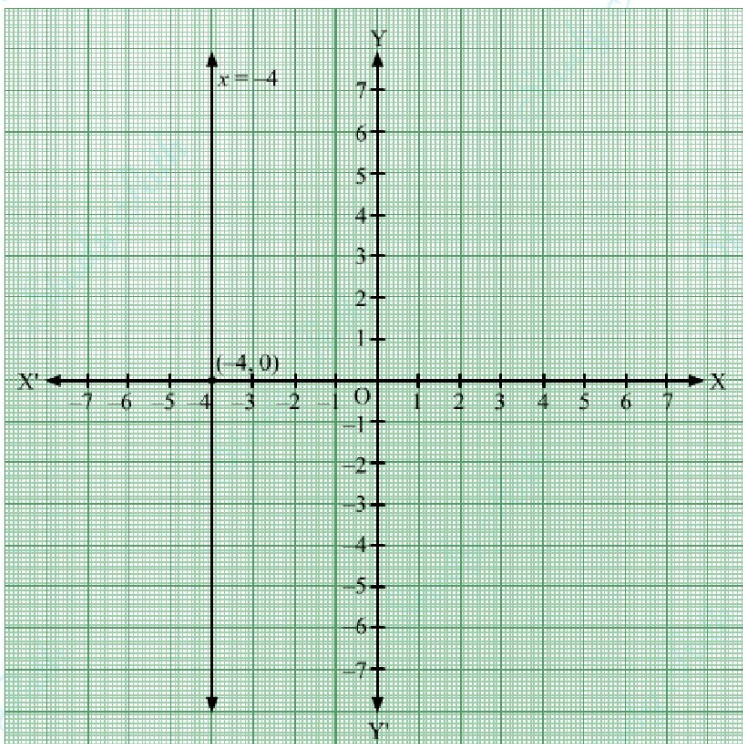
$$\Rightarrow 5x - 2y = 0$$

EXERCISE 4B

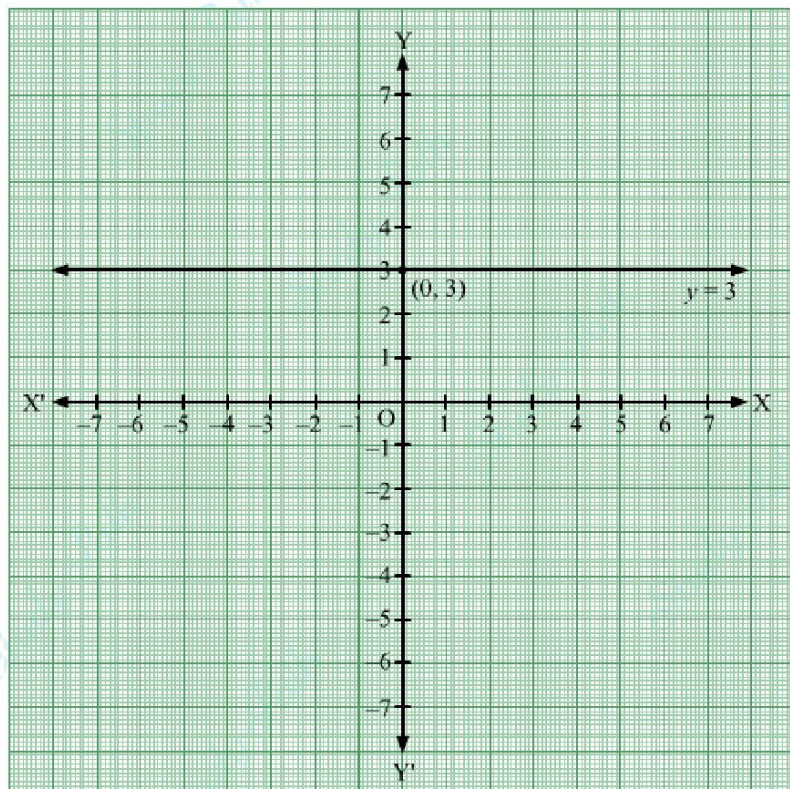
Solution 1: (i) The equation of given line is $x = 4$. This equation does not contain the term of y . So, the graph of this line is parallel to y -axis passing through the point $(4, 0)$.



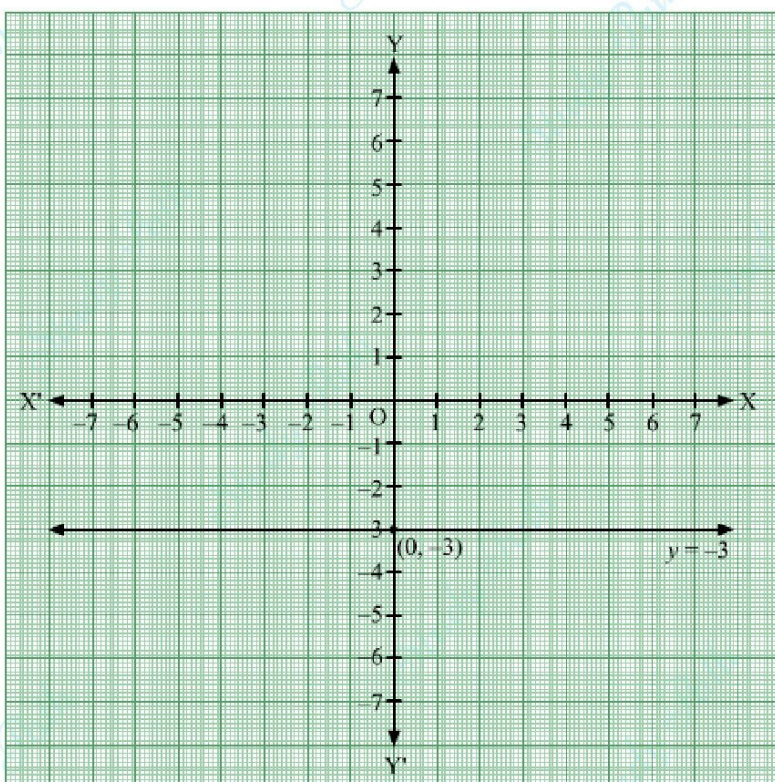
(ii) The equation of given line is $x + 4 = 0$ or $x = -4$. This equation does not contain the term of y . So, the graph of this line is parallel to y -axis passing through the point $(-4, 0)$.



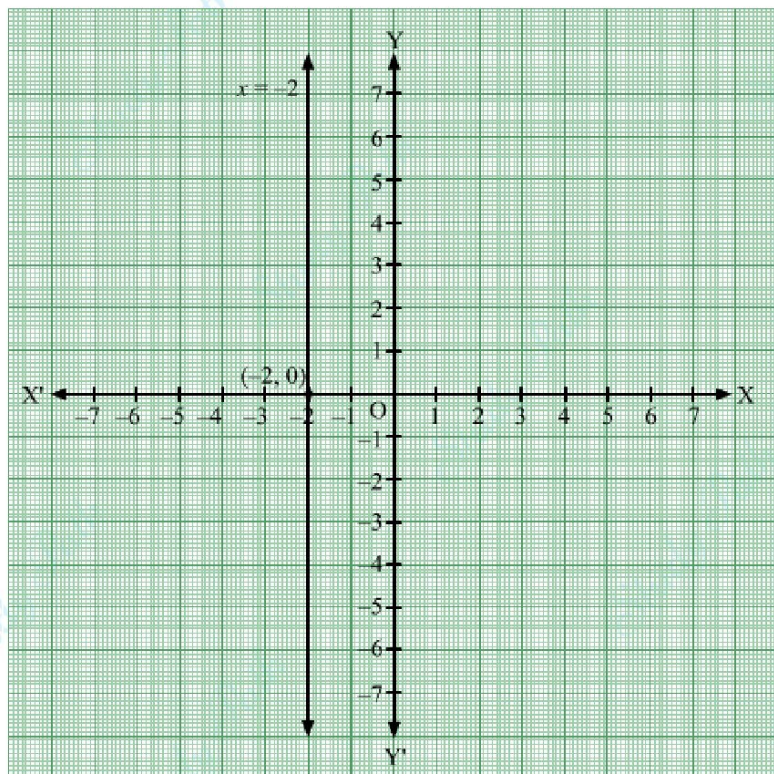
(iii) The equation of given line is $y = 3$. This equation does not contain the term of x . So, the graph of this line is parallel to x -axis passing through the point $(0, 3)$.



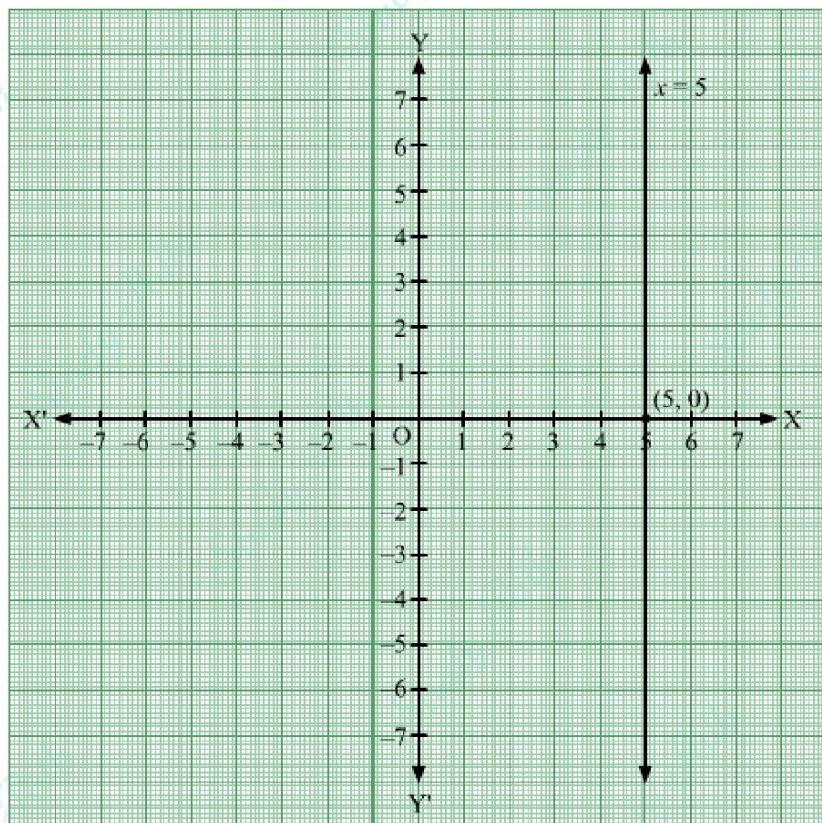
(iv) The equation of given line is $y = -3$. This equation does not contain the term of x . So, the graph of this line is parallel to x -axis passing through the point $(0, -3)$.



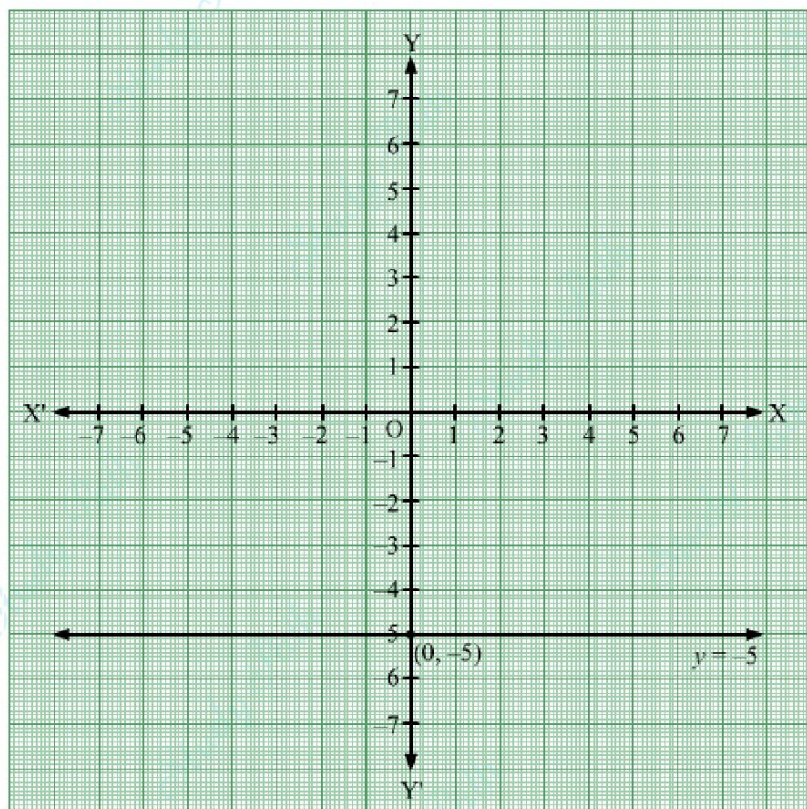
(v) The equation of given line is $x = -2$. This equation does not contain the term of y . So, the graph of this line is parallel to y -axis passing through the point $(-2, 0)$.



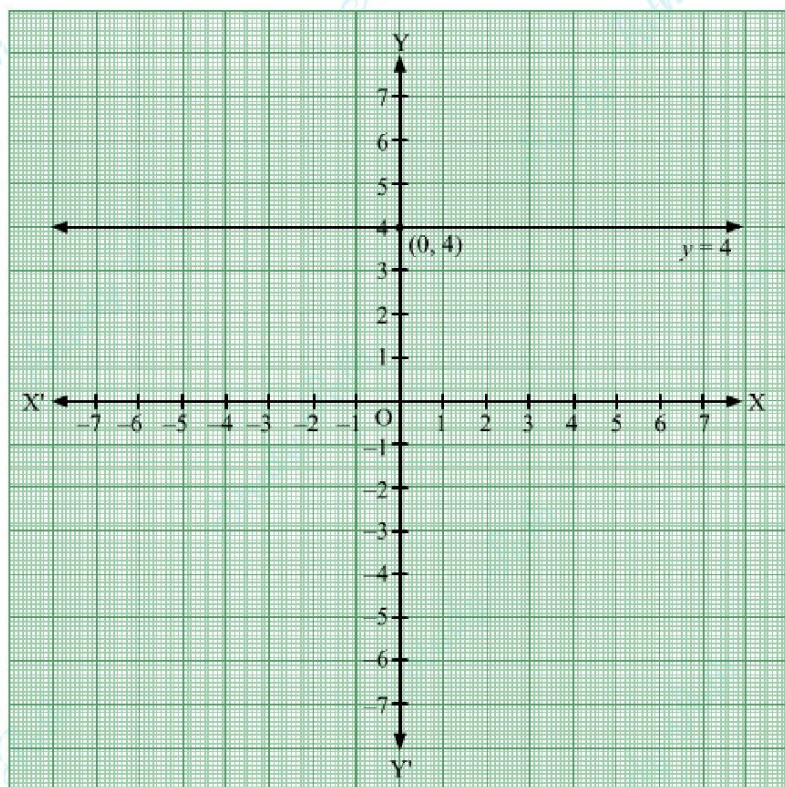
(vi) The equation of given line is $x = 5$. This equation does not contain the term of y . So, the graph of this line is parallel to y -axis passing through the point $(5, 0)$.



(vii) The equation of given line is $y + 5 = 0$ or $y = -5$. This equation does not contain the term of x . So, the graph of this line is parallel to x -axis passing through the point $(0, -5)$.



(viii) The equation of given line is $y = 4$. This equation does not contain the term of x . So, the graph of this line is parallel to x -axis passing through the point $(0, 4)$.



Solution 2:

(i) Given equation: $y = 3x$

When $x = 2$, $y = 6$.

When $x = 1$, $y = 3$.

Thus, we have the following table:

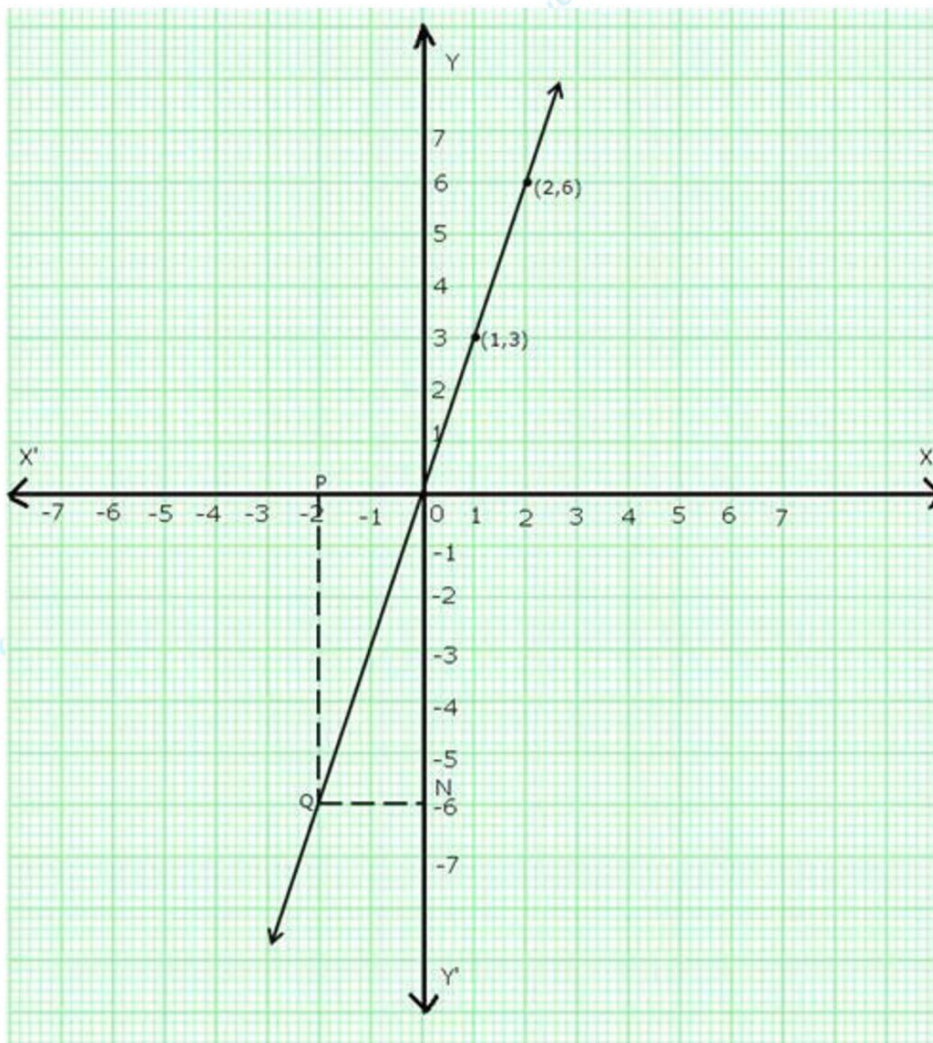
x	2	1
y	6	3

Now plot the points $(2, 6)$, $(1, 3)$ on a graph paper.

Join the points and extend the line in both the directions.

The line segment is the required graph of the equation.

From the graph we can see that when $x = 2$, $y = 6$



(ii) Given equation: $y = 3x$

When $x = -2$, $y = -6$.

When $x = -1$, $y = -3$.

Thus, we have the following table:

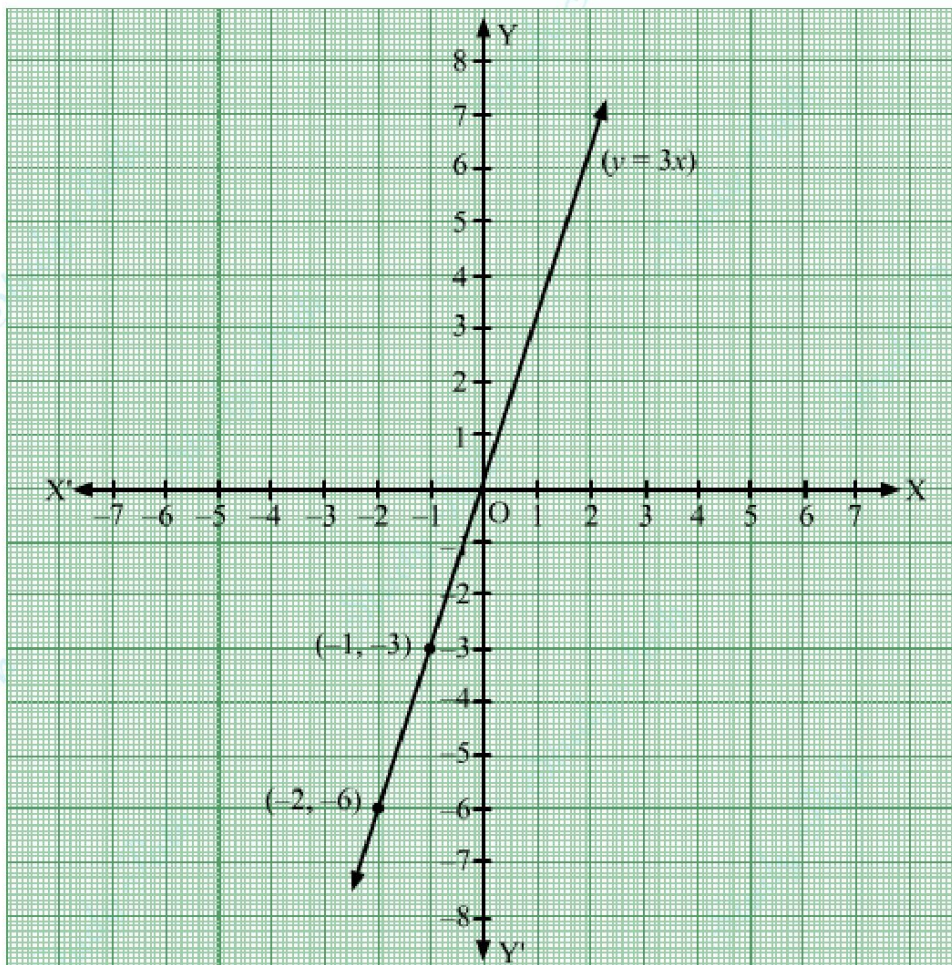
x	-2	-1
y	-6	-3

Now plot the points $(-2, -6)$, $(-1, -3)$ on a graph paper.

Join the points and extend the line in both the directions.

The line segment is the required graph of the equation.

From the graph we can see that when $x = -2$, $y = -6$



Solution 3: Given equation:

$$x + 2y - 3 = 0$$

$$\Rightarrow x + 2y = 3$$

When $y = 0$,

$$x + 0 = 3$$

$$\Rightarrow x = 3$$

When $y = 1$,
 $x + 2 = 3$
 $\Rightarrow x = 3 - 2 = 1$

When $y = 2$,
 $x + 4 = 3$
 $\Rightarrow x = 3 - 4$
 $\Rightarrow x = -1$

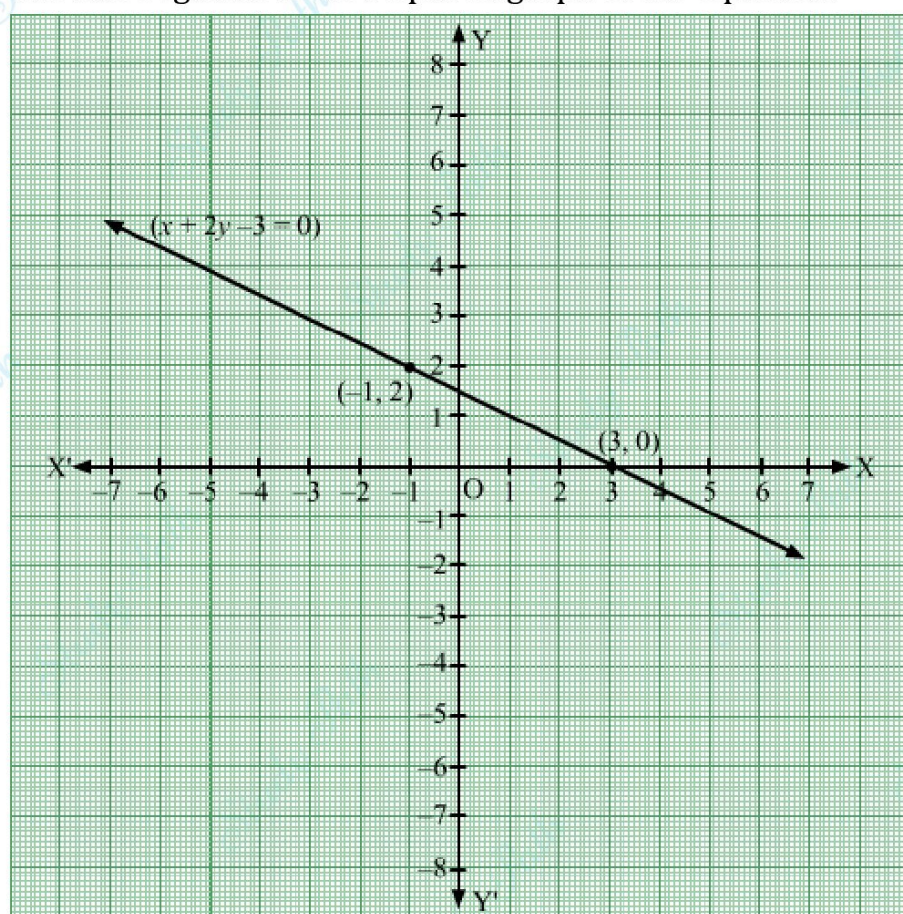
Thus, we have the following table:

x	3	1	-1
y	0	1	2

Now plot the points $(3,0)$, $(1,1)$ and $(-1,2)$ on the graph paper.

Join the points and extend the line in both the directions.

The line segment is the required graph of the equation.



When $x = 5$,
 $y = \frac{3 - x}{2}$

$$\Rightarrow y = \frac{3-5}{2}$$

$$\Rightarrow y = \frac{-2}{2}$$

$$\Rightarrow y = -1$$

Similarly, from the graph we can see that when $x = -5$, $y = 4$.

Solution 4: Given equation:

$$2x - 3y = 5$$

$$\Rightarrow 2x = 3y + 5$$

$$\Rightarrow x = \frac{3y + 5}{2}$$

$$\text{When, } y = -1, x = \frac{-3 + 5}{2} = \frac{2}{2} = 1$$

$$\text{When, } y = -3, x = \frac{-9 + 5}{2} = \frac{-4}{2} = -2$$

Thus, we have the following table:

x	1	-2
y	-1	-3

Plot the points $(-2, -3)$, $(1, -1)$, on the graph paper and extend the line in both directions.

(i) When $x = 4$:

$$4 = \frac{3y + 5}{2}$$

$$\Rightarrow 8 = 3y + 5$$

$$\Rightarrow 3y = 8 - 5$$

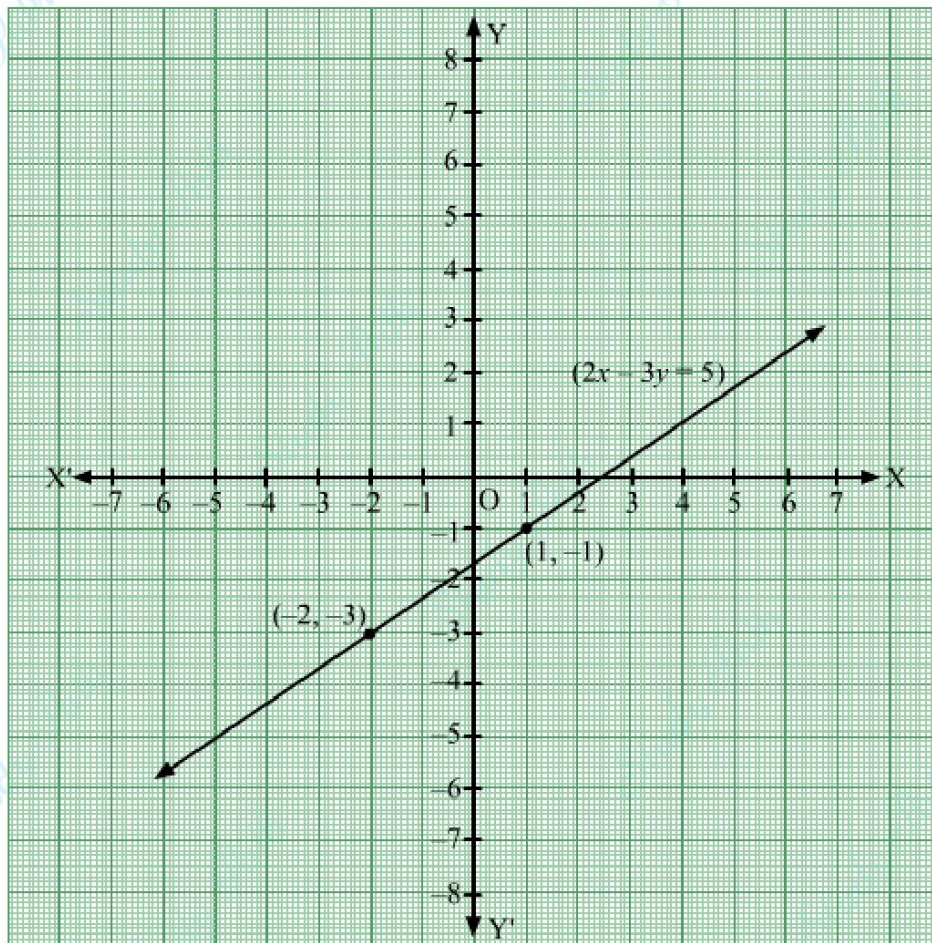
$$\Rightarrow 3y = 3$$

$$\Rightarrow 3y = 3$$

$$\Rightarrow y = 1$$

(ii) When $y = 3$:

$$\begin{aligned}x &= \frac{3y + 5}{2} \\&= \frac{14}{2} \\&= 7\end{aligned}$$



Solution 5: Given equation:

$$\begin{aligned}2x + y &= 6 \\ \Rightarrow y &= 6 - 2x\end{aligned}$$

When, $x = 0$, $y = 6 - 0 = 6$

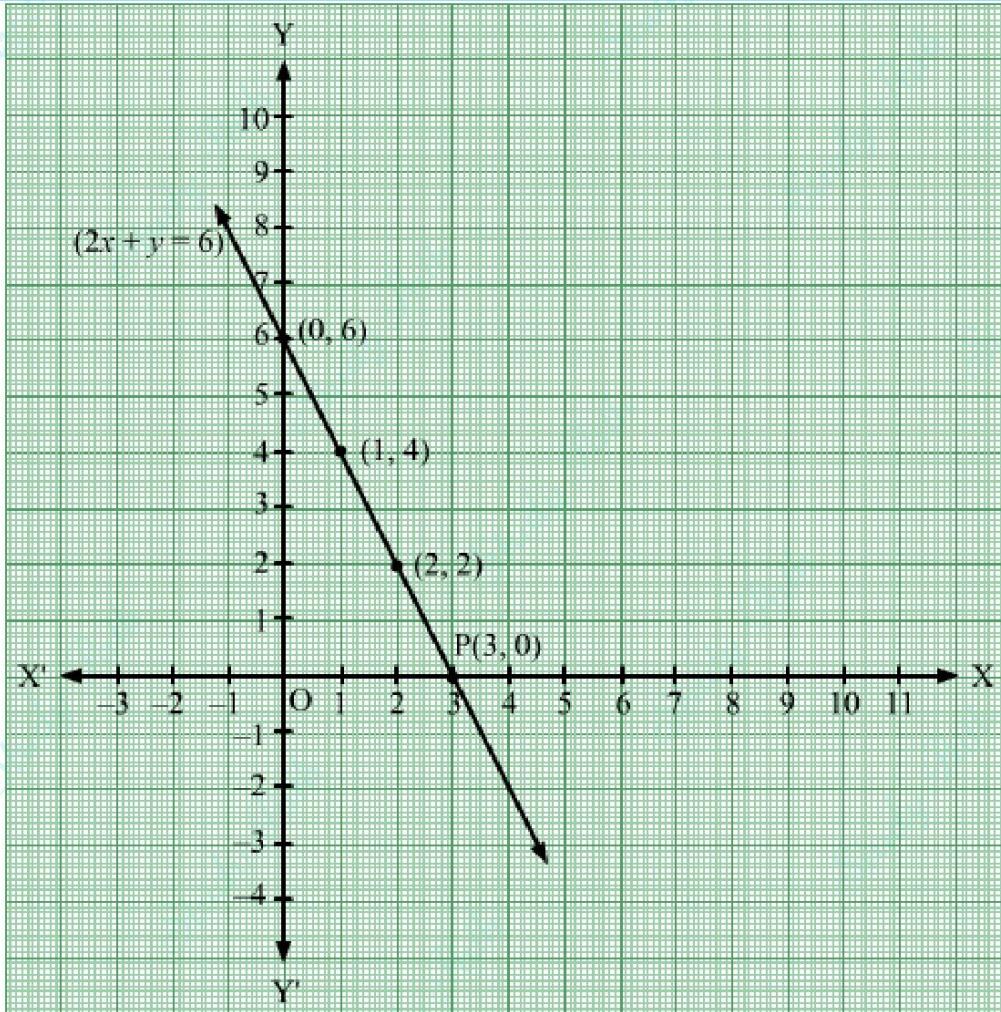
When, $x = 1$, $y = 6 - 2 = 4$

When, $x = 2$, $y = 6 - 4 = 2$

Thus, we have the following table:

xx	0	1	2
yy	6	4	2

Plot the points $(0,6)$, $(1,4)$ and $(2,2)$ on the graph paper. Join these points and extend the line.



Clearly, the graph cuts the x-axis at $P(3, 0)$.

Solution 6: Given equation:

$$3x + 2y = 6.$$

$$\Rightarrow 2y = 6 - 3x$$

$$\Rightarrow y = \frac{6 - 3x}{2}$$

When $x = 2$,

$$y = \frac{6 - 6}{2} = \frac{0}{2} = 0$$

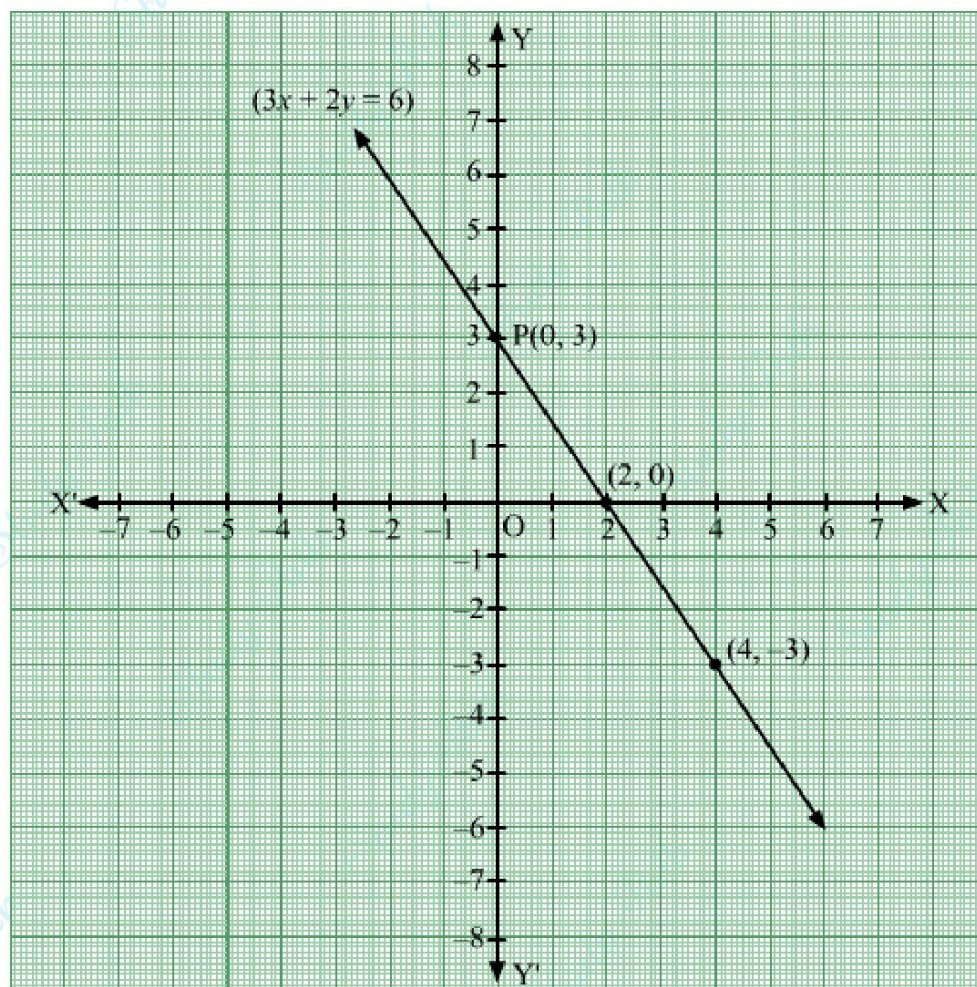
When $x = 4$,

$$y = \frac{6 - 12}{2} = -3$$

Thus, we get the following table:

x	2	4
y	0	-3

Plot the points $(2,0)$, $(4,-3)$ on the graph paper. Join the points and extend the graph in both the directions.



Clearly, the graph cuts the y-axis at $P(0,3)$.

Solution 7:

$$3x - 2y = 4$$

$$\Rightarrow 2y = 3x - 4$$

$$\Rightarrow y = \frac{3x - 4}{2}$$

When $x = 0$,

$$y = \frac{3 \times 0 - 4}{2} = \frac{0 - 4}{2} = -\frac{4}{2} = -2$$

When $x = 2$,

$$y = \frac{3 \times 2 - 4}{2} = \frac{6 - 4}{2} = \frac{2}{2} = 1$$

When $x = -2$,

$$y = \frac{3 \times (-2) - 4}{2} = \frac{-6 - 4}{2} = \frac{-10}{2} = -5$$

Thus, the points on the line $3x - 2y = 4$ are as given in the following table:

x	0	2	-2
y	-2	1	-5

Plotting the points $(0, -2)$, $(2, 1)$ and $(-2, -5)$ and drawing a line passing through these points, we obtain the graph of the line $3x - 2y = 4$.

$$x + y - 3 = 0$$

$$\Rightarrow y = -x + 3$$

When $x = 0$,

$$y = -0 + 3 = 3$$

When $x = 1$,

$$y = -1 + 3 = 2$$

When $x = -1$,

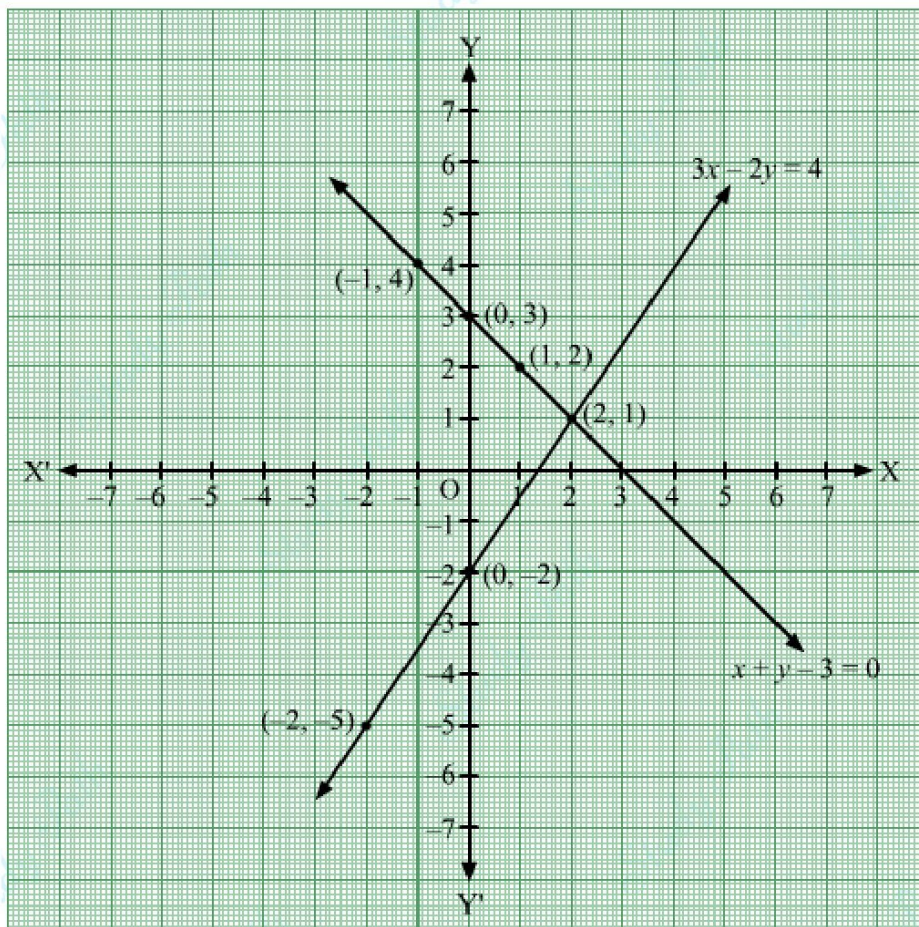
$$y = -(-1) + 3 = 1 + 3 = 4$$

Thus, the points on the line $x + y - 3 = 0$ are as given in the following table:

x	0	1	-1
y	3	2	4

Plotting the points $(0, 3)$, $(1, 2)$ and $(-1, 4)$ and drawing a line passing through these points, we obtain the graph of the line $x + y - 3 = 0$.

It can be seen that the lines $3x - 2y = 4$ and $x + y - 3 = 0$ intersect at the point $(2, 1)$.



Solution 8:

$$4x + 3y = 24$$

$$\Rightarrow 3y = -4x + 24$$

$$\Rightarrow y = \frac{-4x + 24}{3}$$

When $x = 0$,

$$y = \frac{-4 \times 0 + 24}{3} = \frac{0 + 24}{3} = \frac{24}{3} = 8$$

When $x = 3$,

$$y = \frac{-4 \times 3 + 24}{3} = \frac{-12 + 24}{3} = \frac{12}{3} = 4$$

When $x = 6$,

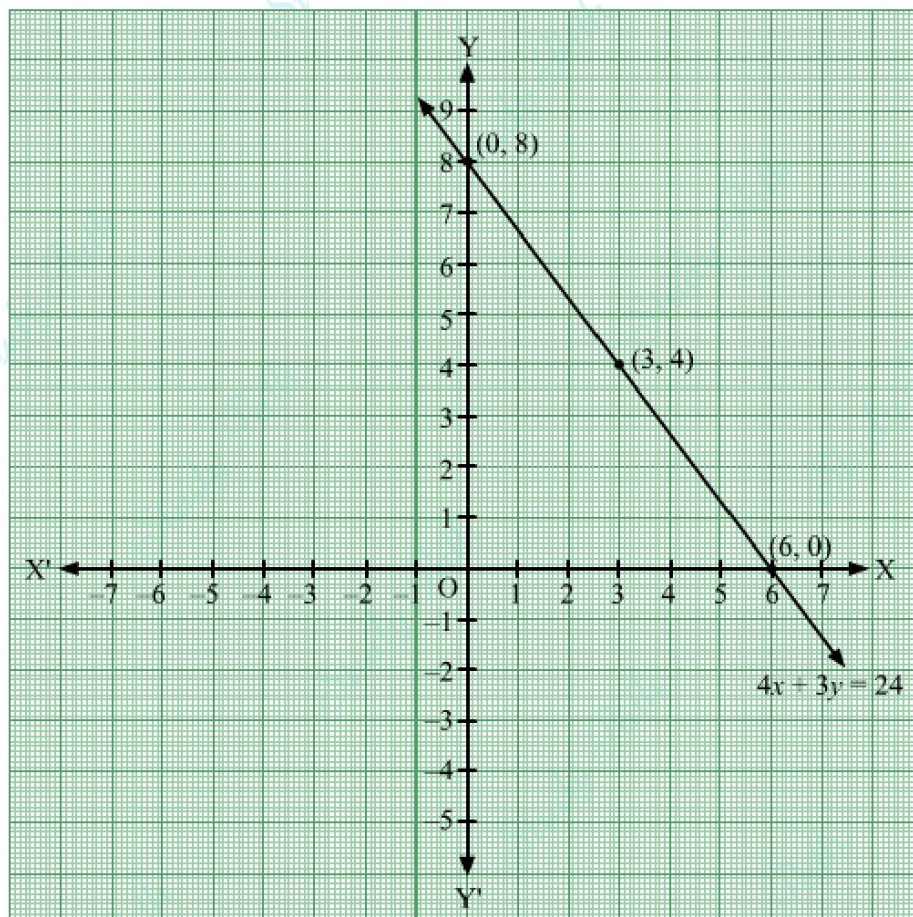
$$y = \frac{-4 \times 6 + 24}{3} = \frac{-24 + 24}{3} = \frac{0}{3} = 0$$

Thus, the points on the line $4x + 3y = 24$ are as given in the following table:

x	0	3	6
y	8	4	0

Plotting the points $(0, 8)$, $(3, 4)$ and $(6, 0)$ and drawing a line passing through these points, we obtain the graph of the line $4x + 3y = 24$.

(i) It can be seen that the line $4x + 3y = 24$ intersects the x-axis at $(6, 0)$ and y-axis at $(0, 8)$.



(ii) The triangle formed by the line and the axes is a right triangle, right-angled at the origin.

$$\therefore \text{Area of the triangle} = \frac{1}{2} \times 6 \times 8 = 24 \text{ square units}$$

Solution 9:

$$2x + y = 6$$

$$\Rightarrow y = -2x + 6$$

When $x = 0$,

$$y = -2 \times 0 + 6 = 0 + 6 = 6$$

When $x = 1$,

$$y = -2 \times 1 + 6 = -2 + 6 = 4$$

When $x = 2$,

$$y = -2 \times 2 + 6 = -4 + 6 = 2$$

Thus, the points on the line $2x + y = 6$ are as given in the following table:

x	0	1	2
y	6	4	2

Plotting the points $(0, 6)$, $(1, 4)$ and $(2, 2)$ and drawing a line passing through these points, we obtain the graph of the line $2x + y = 6$.

$$2x - y + 2 = 0$$

$$\Rightarrow y = 2x + 2$$

When $x = 0$,

$$y = 2 \times 0 + 2 = 0 + 2 = 2$$

When $x = 1$,

$$y = 2 \times 1 + 2 = 2 + 2 = 4$$

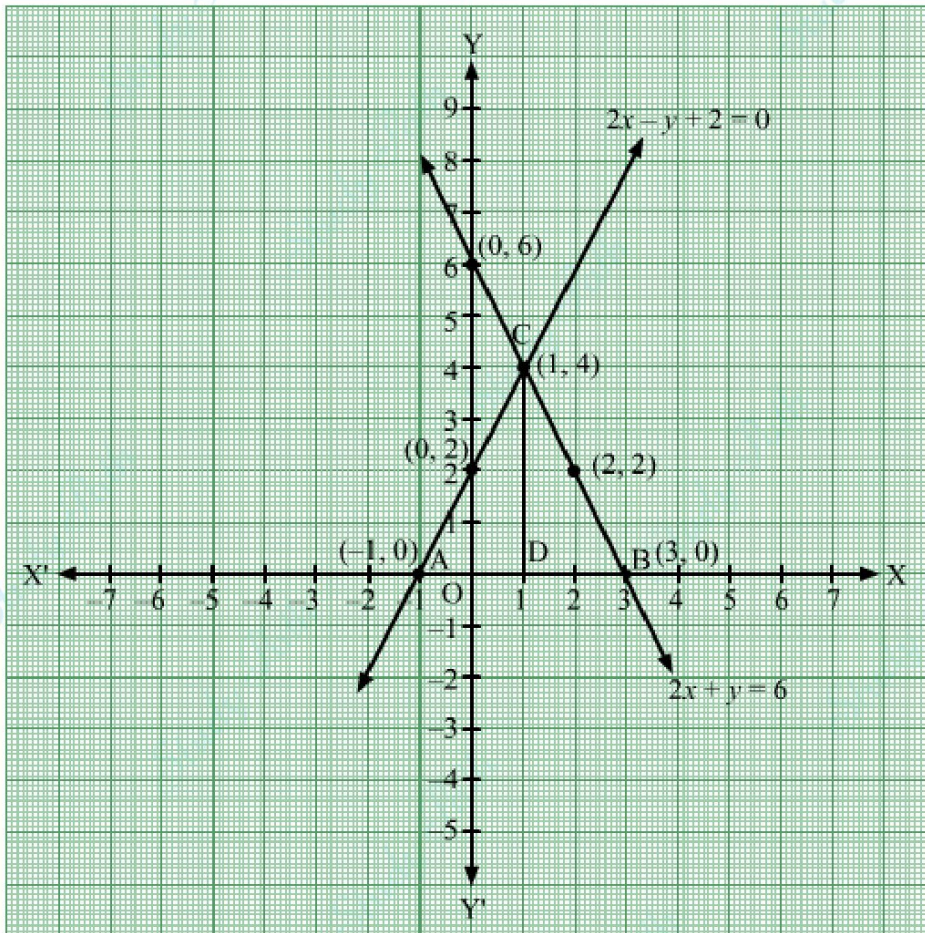
When $x = -1$,

$$y = 2 \times (-1) + 2 = -2 + 2 = 0$$

Thus, the points on the line $2x - y + 2 = 0$ are as given in the following table:

x	0	1	-1
y	2	4	0

Plotting the points $(0, 2)$, $(1, 4)$ and $(-1, 0)$ and drawing a line passing through these points, we obtain the graph of the line $2x - y + 2 = 0$.



The shaded region represents the area bounded by the lines $2x + y = 6$, $2x - y + 2 = 0$ and the x-axis. This represents a triangle.

It can be seen that the lines intersect at the point $C(1, 4)$. Draw CD perpendicular from C on the x-axis.

Height = $CD = 4$ units

Base = $AB = 4$ units

$\therefore$ Area of the shaded region

= Area of $\triangle ABC$

= $\frac{1}{2} \times AB \times CD$

= $\frac{1}{2} \times 4 \times 4$

= 8 square units

Solution 10:

$$x - y = 1$$

$$\Rightarrow y = x - 1$$

When $x = 0$,

$$y = 0 - 1 = -1$$

When $x = 1$,

$$y = 1 - 1 = 0$$

When $x = 2$,

$$y = 2 - 1 = 1$$

Thus, the points on the line $x - y = 1$ are as given in the following table:

x	0	1	2
y	-1	0	1

Plotting the points $(0, -1)$, $(1, 0)$ and $(2, 1)$ and drawing a line passing through these points, we obtain the graph of the line $x - y = 1$.

$$2x + y = 8$$

$$\Rightarrow y = -2x + 8$$

When $x = 1$,

$$y = -2 \times 1 + 8 = -2 + 8 = 6$$

When $x = 2$,

$$y = -2 \times 2 + 8 = -4 + 8 = 4$$

When $x = 3$,

$$y = -2 \times 3 + 8 = -6 + 8 = 2$$

Thus, the points on the line $2x + y = 8$ are as given in the following table:

x	1	2	3
y	6	4	2

$$= \frac{1}{2} \times 9 \times 3$$

$$= \frac{27}{2} \text{ square units}$$

Solution 11:

$$x + y = 6$$

$$\Rightarrow y = -x + 6$$

When $x = 0$,

$$y = -0 + 6 = 6$$

When $x = 1$,

$$y = -1 + 6 = 5$$

When $x = 3$,

$$y = -3 + 6 = 3$$

Thus, the points on the line $x + y = 6$ are as given in the following table:

x	0	1	3
y	6	5	3

Plotting the points $(0, 6)$, $(1, 5)$ and $(3, 3)$ and drawing a line passing through these points, we obtain the graph of the line $x + y = 6$.

$$x - y = 2$$

$$\Rightarrow y = x - 2$$

When $x = 0$,

$$y = 0 - 2 = -2$$

When $x = 2$,

$$y = 2 - 2 = 0$$

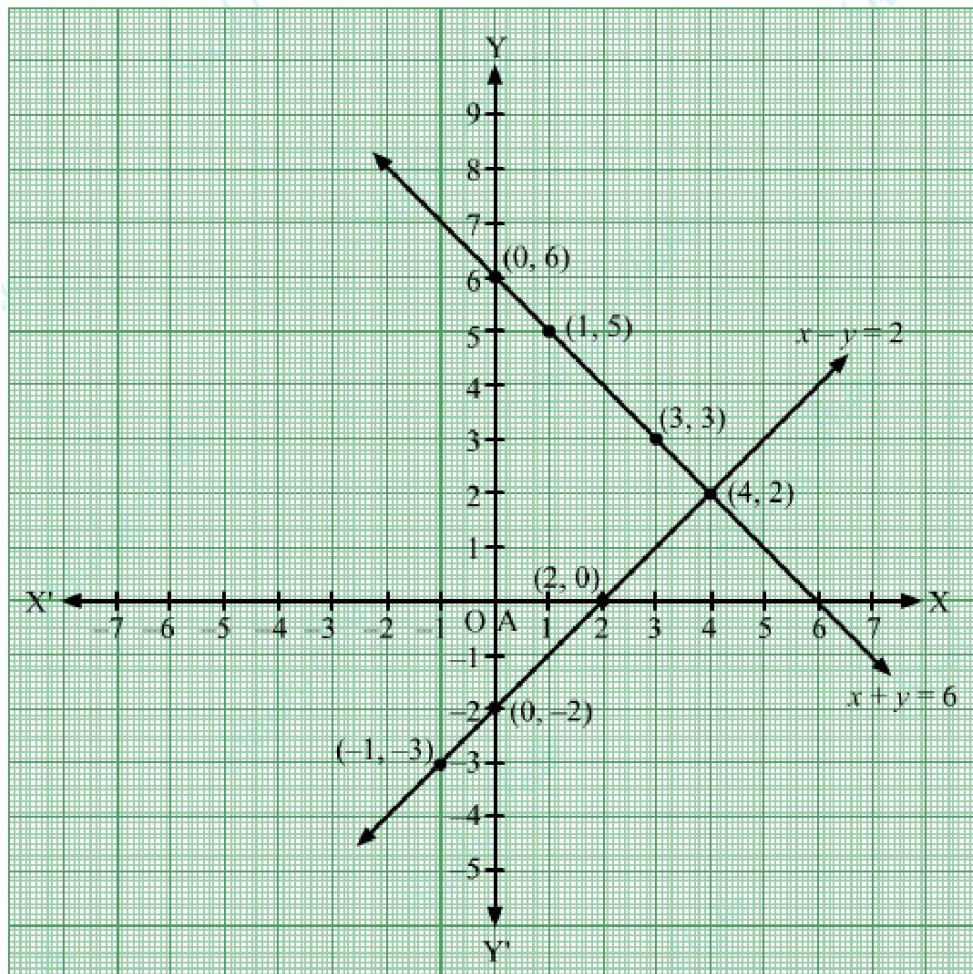
When $x = -1$,

$$y = -1 - 2 = -3$$

Thus, the points on the line $x - y = 2$ are as given in the following table:

x	0	2	-1
y	-2	0	-3

Plotting the points $(0, -2)$, $(2, 0)$ and $(-1, -3)$ and drawing a line passing through these points, we obtain the graph of the line $x - y = 2$.



It can be seen that the lines $x + y = 6$ and $x - y = 2$ intersect at the point $(4, 2)$.

Solution 12: Let the contribution of A and B be ₹ x and ₹ y , respectively.

Total contribution of A and B = ₹ $x + ₹ y = ₹ (x + y)$

It is given that the total contribution of A and B is ₹ 100.

$$\therefore x + y = 100$$

This is the linear equation satisfying the given data.

$$x + y = 100$$

$$\Rightarrow y = 100 - x$$

When $x = 10$,
 $y = 100 - 10 = 90$

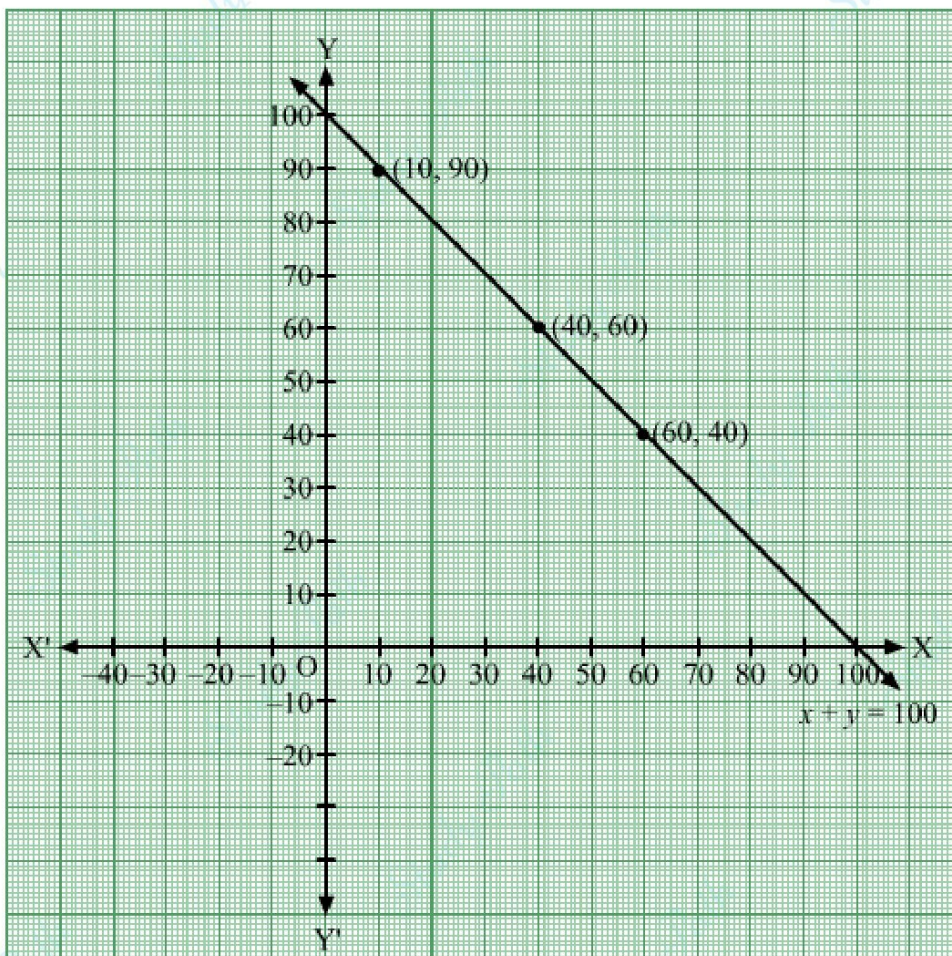
When $x = 40$,
 $y = 100 - 40 = 60$

When $x = 60$,
 $y = 100 - 60 = 40$

Thus, the points on the line $x + y = 100$ are as given in the following table:

x	10	40	60
y	90	60	40

Plotting the points $(10, 90)$, $(40, 60)$ and $(60, 40)$ and drawing a line passing through these points, we obtain the graph of the line $x + y = 100$.



Multiple Choice Questions (MCQ)

Solution 1: (b) $y = 0$

Since, the y-coordinate of any point on x-axis is always 0.

So, the equation of the x-axis is $y = 0$.

Hence, the correct option is (b).

Solution 2: (a) $x = 0$

Since, the x-coordinate of any point on y-axis is always 0.

So, the equation of the y-axis is $x = 0$.

Hence, the correct option is (a).

Solution 3: (c) the line $y = x$

Given, a point of the form (a, a) , where $a \neq 0$.

When $a = 1$, the point is $(1, 1)$

When $a = 2$, the point is $(2, 2)$and so on.

Plot the points $(1, 1)$ and $(2, 2)$and so on. Join the points and extend them in both the direction. You will get the equation of the line $y = x$

Solution 4: (d) the line $x + y = 0$

Given, a point of the form $(a, -a)$, where $a \neq 0$.

When $a = 1$, the point is $(1, -1)$.

When $a = 2$, the point is $(2, -2)$.

When $a = 3$, the point is $(3, -3)$ and so on.

Plot these points on a graph paper. Join these points and extend them in both the directions.

You will get the equation of the line $x + y = 0$

Solution 5: (c) infinitely many solutions

Given linear equation:

$$3x - 5y = 153$$

$$\Rightarrow x = 5y + 153$$

When $y = 0$,

$$x = \frac{15}{3} = 5$$

When $y = 3$,

$$x = \frac{30}{3} = 10$$

When $y = -3$,

$$x = \frac{0}{3} = 0$$

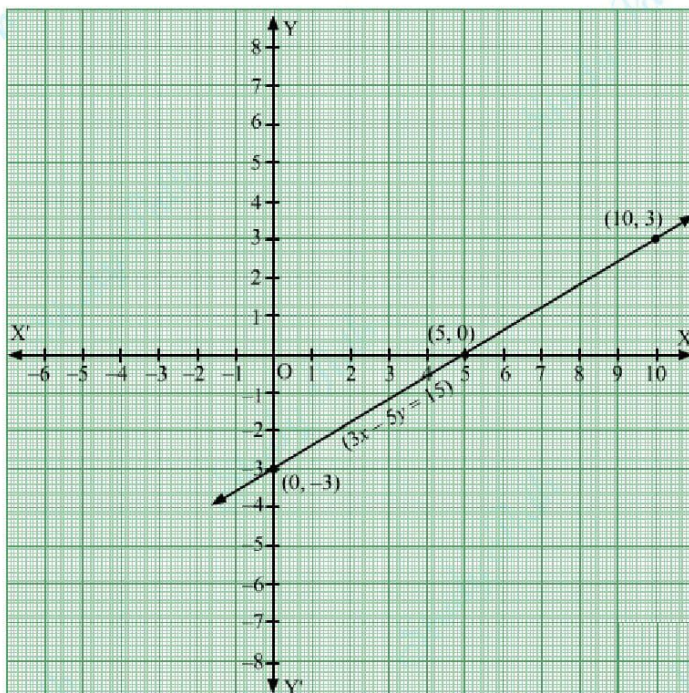
Thus, we have the following table:

x	5	10	0
y	0	3	-3

Plot the points A(5,0) , B(10,3) and C(0,-3). Join the points and extend them in both the directions.

We get infinite points that satisfy the given equation.

Hence, the linear equation has infinitely many solutions.



Solution 6: (a) natural numbers

Since, every point on the line represented by the equation $2x + 5y = 7$ is its solution.

Therefore, there are infinite solutions of the equation the equation $2x + 5y = 7$ in which the values of x and y are rational numbers, positive real numbers or real numbers.

But, as $2 + 5 = 7$, i.e. $x = 1$ and $y = 1$ are the only pair of natural numbers that are the solution of the equation the equation $2x + 5y = 7$.

So, the equation $2x + 5y = 7$ has a unique solution, if x and y are both are natural numbers.

Hence, the correct option is (a).

Solution 7: (c) parallel to the x -axis at a distance of 5 units from the origin

As, the graph of $y = 5$ is a line parallel to x -axis i.e. $y = 0$.

The line represented by the equation $y = 5$ is parallel to x -axis and intersects y -axis at $y = 5$.

So, the graph of $y = 5$ is a line parallel to the x -axis at a distance of 5 units from the origin making an intercept 5 on the y -axis.

Hence, the correct answers are options (b) and (c).

Disclaimer: In this question, there are two correct answers.

Solution 8: (d) parallel to the y -axis at a distance of 4 units from the origin

As, the graph of $x = 4$ is a line parallel to y -axis i.e. $x = 0$.

The line represented by the equation $x = 4$ is parallel to y -axis and intersects x -axis at $x = 4$.

So, the graph of $x = 4$ is parallel to y -axis at a distance of 4 units from the origin making an intercept 4 on the x -axis.

Hence, the correct options are (a) and (d).

Disclaimer: In this question, there are two correct answers.

Solution 9: (c) parallel to the y-axis at a distance of 3 units to the left of y-axis

As, the graph of $x + 3 = 0$ or $x = -3$ is a line parallel to y-axis i.e. $x = 0$.

The line represented by the equation $x = -3$ is parallel to y-axis and intersects x-axis at $x = -3$.

So, the graph of $x + 3 = 0$ is a line parallel to the y-axis at a distance of 3 units to the left of y-axis making an intercept -3 on the x-axis.

Hence, the correct options are (a) and (c).

Disclaimer: In this question, there are two correct answers.

Solution 10: (c) parallel to the x-axis at a distance of 2 units below the x-axis

As, the graph of $y + 2 = 0$ or $y = -2$ is a line parallel to x-axis i.e. $y = 0$.

The line represented by the equation $y = -2$ is parallel to x-axis and intersects y-axis at $y = -2$.

So, the graph of $y + 2 = 0$ is a line parallel to the x-axis at a distance of 2 units below the x-axis making an intercept -2 on the y-axis.

Hence, the correct options are (b) and (c).

Disclaimer: In this question, there are two correct answers.

Solution 11: (c) (0, 2)

When a graph meets the y-axis, the x coordinate is zero.

Thus, substituting $x = 0$ in the given equation, we get

$$2(0) + 3y = 6$$

$$\Rightarrow 3y = 6$$

$$\Rightarrow y = 2$$

So, the point of meeting is (0, 2).

Hence, the correct answer is option (c).

Solution 12: (c) (5, 0)

When a graph meets the x-axis, the y coordinate is zero.

Thus, substituting $y = 0$ in the given equation, we get

$$2x + 5(0) = 10$$

$$\Rightarrow 2x = 10$$

$$\Rightarrow x = 5$$

So, the point of meeting is $(5, 0)$.

Hence, the correct option is (c).

Solution 13: (c) $(3, 2)$

The graph of line $x = 3$ is a line parallel to the y-axis.

Hence, it passes through $(3, 2)$, satisfying $x = 3$.

Solution 14: (c) $(2, 3)$

Since, the graph of the line $y = 3$ is parallel to x-axis at a distance of 3 units from the x-axis.

Or, the y-coordinate of every point on the line is always equal to 3.

So, the graph of the line $y = 3$ passes through the point $(2, 3)$.

Hence, the correct option is (c).

Solution 15: (d) $(-3, 2)$

The graph of the line $y = -3$ does not pass through $(-3, 2)$ since $(-3, 2)$ does not satisfy $y = -3$.

Solution 16: (d) $(1, 1)$

Given equation:

$$x - y = 0$$

$$\text{or, } x = y$$

$$\text{When } x = 1, y = 1$$

$$\text{When } x = 2, y = 2 \dots \text{and so on}$$

Thus, we get the following table:

x	1	2
y	1	2

Plot the points on the graph paper. Join the points and extend them in both the directions.

We can see the linear equation $x - y = 0$ passes through the point (1,1).

Solution 16: (b) $x + y = 0$

Given points: $(-2, 2)$, $(0, 0)$ and $(2, -2)$

We have to check which equation satisfies the given points.

Let us check for (a) $x - y = 0$ $x - y = 0$

Substituting $(-2, 2)$ in the equation, we get: $x - y = -2 - 2 = -4$

Substituting $(0, 0)$ in the equation, we get: $x - y = 0 - 0 = 0$

Substituting $(2, -2)$ in the equation, we get: $x - y = 2 + 2 = 4$

So, the given points do not satisfy the equation.

Now, let us check (b) $x + y = 0$ $x + y = 0$

Substituting $(-2, 2)$ in the equation, we get: $x + y = -2 + 2 = 0$

Substituting $(0, 0)$ in the equation, we get: $x + y = 0 + 0 = 0$

Substituting $(2, -2)$ in the equation, we get: $x + y = 2 - 2 = 0$

So, the given points satisfy the equation.

Similarly, we can check other equations and find that the given points will not satisfy the equations.

Hence, each of $(-2, 2)$, $(0, 0)$ and $(2, -2)$ is a solution of the linear equation $x + y = 0$.

Solution 18: (c) Infinitely many

Infinite linear equations are satisfied by $x = 2, y = 3$

Solution 19: (a) $a \neq 0, b \neq 0$

A linear equation in two variables $ax + by + c = 0$, where $a \neq 0$ and $b \neq 0$.

Solution 20: (d) 4

Since, $(2, 0)$ is a solution of the linear equation $2x + 3y = k$, substituting $x = 2$ and $y = 0$ in the given equation, we have

$$2(2) + 3(0) = k$$

$$\Rightarrow 4 + 0 = k$$

$$\Rightarrow k = 4$$

Hence, the correct option is (d).

Solution 21: (c) $(x, 0)$,

Any point on the x -axis is of the form $(x, 0)$, where $x \neq 0$.

Solution 22: (b) $(0, y)$,

Any point on the y -axis is of the form $(0, y)$, where $y \neq 0$.

Solution 23: (c) $x + y = 7$

Substituting $x = 5$ and $y = 2$ in L.H.S. of equation $x + y = 7$, we get

$$\text{L.H.S.} = 5 + 2 = 7 = \text{R.H.S.}$$

Hence, $x = 5$ and $y = 2$ is a solution of the linear equation $x + y = 7$.

Solution 24: (b) $\frac{5}{3}$

Since the point $(3, 4)$ lies on the graph of $3y = ax + 7$, substituting $x = 3$ and $y = 4$ in the given equation, we get

$$3(4) = a(3) + 7$$

$$\Rightarrow 12 = 3a + 7$$

$$\Rightarrow 3a = 5$$

$$\Rightarrow a = \frac{5}{3}$$